
MA 644: Dynamical Systems

Course Code	MA 644
Course Name	Dynamical Systems
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	Ph.D./ M.S./B.Tech
Distribution	Elective
Approval	1st Adhoc Senate (Apr 2011)

Course Contents

Dynamical systems- Central manifold and Normal form, Plane Autonomous Systems, Attractors, Map-1D map, Logistic map, Poincare' maps, generalized Baker's map, circle map, Bifurcations-. Necessary Condition for bifurcation, Saddle node bifurcations, Transcritical Bifurcation, Pitchfork Bifurcation, Normal. forms of. different types of bifurcations, Reduction to Normal Form, Hopf bifurcation, Strange attractors, Henoh map and Rossler system, Box-counting, pointwise and correlation, Hausdorff dimensions, Lyapunov exponent, chaotic transitions, intermittency, crisis.

Text Books

1. Guckenheimer, J., and P. Holmes, **Nonlinear Oscillations, Dynamical Systems and Bifurcations of Vector Fields**, Springer-Verlag, 2002.

References

1. Drazin, P. G., **Nonlinear systems**, Cambridge University Press, 1992.
2. Peitgen, H. O., H. Jurgens, and D. Saupe. **Chaos and Fractals: New Frontiers of Science**, Springer, 2004.
3. Jordan, D. W., and P. Smith, **Nonlinear Ordinary Differential Equations**, Oxford University Press, 1999.
4. Berge, P., Y. Pomeau, and C. Vidal. **Order Within Chaos**, Wiley.